

Doc 9: SVD, Low-Rank, and PCA

Module 1: Linear algebra | Part 1 of 3: Learn (theory)

1. The idea: every matrix is rotate, stretch, rotate

Any matrix, square or not, can be written as $\mathbf{A} = \mathbf{U} \Sigma \mathbf{V}^T$. Reading right to left (Doc 7): $\mathbf{V}^T$ turns the input, Σ stretches along the axes by the **singular values** $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$, and $\mathbf{U}$ turns the result. Picture: $\mathbf{A}$ sends the unit circle to an ellipse, and the singular values are the ellipse's semi-axes.

$$\mathbf{A} = \mathbf{U} \Sigma \mathbf{V}^T \text{ and } \mathbf{A} \mathbf{v}_i = \sigma_i \mathbf{u}_i$$

$\mathbf{U}$ and $\mathbf{V}$ are **orthonormal**: their columns have length 1 and are perpendicular (dot product 0). So $\mathbf{U}^T \mathbf{U} = \mathbf{I}$ and $\mathbf{V}^T \mathbf{V} = \mathbf{I}$, which means $\mathbf{U}^{-1} = \mathbf{U}^T$: turning back is easy. Σ is diagonal and holds the singular values (inverting a diagonal matrix is easy: Doc 6).

2. Computing a 2x2 SVD by hand

- **Step 1.** Compute $\mathbf{A}^T \mathbf{A}$ (a symmetric matrix, Doc 5).
- **Step 2.** Find its eigenvalues $\lambda_1 \geq \lambda_2$ (Doc 8). The singular values are $\sigma_i = \text{sqrt}(\lambda_i)$.
- **Step 3.** Find its eigenvectors and scale each to length 1. They are the columns of $\mathbf{V}$.
- **Step 4.** Get the columns of $\mathbf{U}$ from $\mathbf{u}_i = \mathbf{A} \mathbf{v}_i / \sigma_i$.
- **Step 5.** Check that $\mathbf{U} \Sigma \mathbf{V}^T$ gives back $\mathbf{A}$.

Why it works: $\mathbf{A}^T \mathbf{A} = \mathbf{V} \Sigma^2 \mathbf{V}^T$, so $\mathbf{V}$ holds its eigenvectors and Σ^2 its eigenvalues. A pair $(\mathbf{u}_i, \mathbf{v}_i)$ may flip sign together; the product is unchanged.

3. Facts to remember

Singular values	$\sigma \geq 0$, listed from largest to smallest	never negative
Rank	= the number of nonzero singular values	$\sigma = 0$ means a direction is squashed
Link to eigenvalues	σ_i^2 = eigenvalues of $\mathbf{A}^T \mathbf{A}$	and also of $\mathbf{A} \mathbf{A}^T$
Determinant	$ \det \mathbf{A} = \sigma_1 \sigma_2$ (square $\mathbf{A}$)	the area-scaling factor (Doc 6)
Sum of rank-1 pieces	$\mathbf{A} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots$	each piece is an outer product (Doc 5)
Symmetric matrix	singular values = eigenvalues	a negative λ gives a sign flip in $\mathbf{U}$ or $\mathbf{V}$

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4. Low-rank approximation: keep only the biggest pieces

Keep the k largest terms: $\mathbf{A}_k = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \dots + \sigma_k \mathbf{u}_k \mathbf{v}_k^T$. This is the **best rank- k approximation** of $\mathbf{A}$ (Eckart-Young theorem): no other rank- k matrix is closer.

Error	$\ \mathbf{A} - \mathbf{A}_k\ = \sqrt{\sigma_{k+1}^2 + \sigma_{k+2}^2 + \dots}$	the discarded singular values
Energy kept	$(\sigma_1^2 + \dots + \sigma_k^2) / (\text{all } \sigma^2)$	e.g. $144 / 160 = 90\%$
Storage	$k(m + n + 1)$ numbers instead of $m \cdot n$	rank-10 for 100×100 : 2,010 instead of 10,000

$\|\cdot\|$ is the Frobenius norm: square every entry, add, take the square root.

5. PCA: find the directions where the data varies most

Data matrix $\mathbf{X}$ has n rows (samples) and d columns (features).

- **Step 1. Center:** subtract each column's mean, giving $\mathbf{X}_c$ (Doc 3).
- **Step 2. Covariance:** $\mathbf{C} = \mathbf{X}_c^T \mathbf{X}_c / (n - 1)$, a $d \times d$ symmetric matrix.
- **Step 3. Eigen-decomposition (Doc 8):** the eigenvectors of $\mathbf{C}$ are the **principal directions**; the eigenvalues are the **variances** along them.
- **Step 4. Project:** the scores are $\mathbf{X}_c \mathbf{v}$, using the top k directions $\mathbf{v}$.
- **Step 5. Explained variance:** $\lambda_i / (\lambda_1 + \lambda_2 + \dots)$.

PCA is the SVD of the centered data: the columns of $\mathbf{V}$ are the principal directions, and $\lambda_i = \sigma_i^2 / (n - 1)$. Centering matters: without it the first direction points at the mean instead of at the spread.

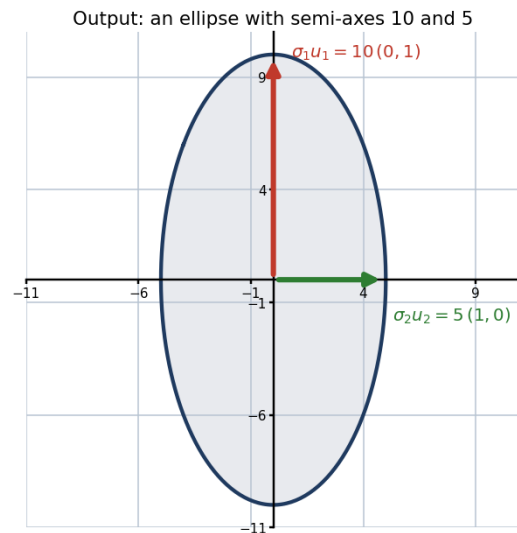
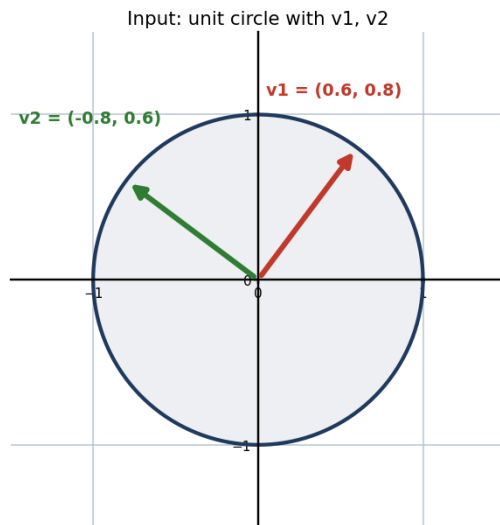
6. Where machine learning uses these

- **Compression and denoising:** keep the top singular values of an image or a data matrix, discard the rest.
- **Dimensionality reduction:** PCA shrinks 1,000 features to 20 for faster models and 2D plots.
- **Recommenders:** a users $\times$ items table is approximated by (users $\times k$)($k \times$ items): matrix factorization.
- **LoRA** (Doc 31) fine-tunes a big weight matrix by adding a low-rank product $\mathbf{B} \mathbf{A}$ instead of changing every entry.
- **Embeddings:** a low-rank view of a huge word-context table is one way to build word vectors.

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Part 2 of 3: Worked examples. Every cell filled in. Read, then cover the answers and redo them.

Example 1: The SVD of $A = \begin{bmatrix} -4 & 3 \\ 6 & 8 \end{bmatrix}$ (every number comes out exact)



Step 1: $A^T A$

		-4	3
		6	8
-4	6	52	36
3	8	36	73

Step 2. trace = $52 + 73 = 125$, det = $52 \times 73 - 36 \times 36 = 3796 - 1296 = 2500$. $\lambda^2 - 125\lambda + 2500 = 0$, so $(\lambda - 100)(\lambda - 25) = 0$: $\lambda = 100, 25$, so $\sigma_1 = 10, \sigma_2 = 5$.

Step 3. $\lambda = 100$: $[-48 \ 36; 36 \ -27]$ gives $-48x + 36y = 0$, direction $(3, 4)$, length 5, so $v_1 = (0.6, 0.8)$. $\lambda = 25$: $[27 \ 36; 36 \ 48]$ gives $27x + 36y = 0$, direction $(-4, 3)$, so $v_2 = (-0.8, 0.6)$.

Step 4. $u_1 = A v_1 / 10 = (0, 10) / 10 = (0, 1)$. $u_2 = A v_2 / 5 = (5, 0) / 5 = (1, 0)$.

Step 5. Check. $U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\Sigma = \text{diag}(10, 5)$, $V^T = \begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix}$:

U times Σ

		10	0
		0	5
0	1	0	5
1	0	10	0

$(U \Sigma)$ times V^T = A

		0.6	0.8
		-0.8	0.6
0	5	-4	3
10	0	6	8

The product returns $A = \begin{bmatrix} -4 & 3 \\ 6 & 8 \end{bmatrix}$. The circle's direction v_1 lands on the long axis, stretched by 10.

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Part 2 of 3: Worked examples, continued

Example 2: A matrix as a sum of rank-1 pieces (a 4x4 checkerboard)

Here $\sigma_1 = 12$, $\sigma_2 = 4$, $u_1 = v_1 = \frac{1}{2}(1, 1, 1, 1)$, $u_2 = \frac{1}{2}(1, 1, -1, -1)$, $v_2 = \frac{1}{2}(1, -1, 1, -1)$. Each piece $\sigma u v^T$ is an outer product (Doc 5).

$$\begin{array}{|c|c|c|c|} \hline 4 & 2 & 4 & 2 \\ \hline 4 & 2 & 4 & 2 \\ \hline 2 & 4 & 2 & 4 \\ \hline 2 & 4 & 2 & 4 \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline 3 & 3 & 3 & 3 \\ \hline 3 & 3 & 3 & 3 \\ \hline 3 & 3 & 3 & 3 \\ \hline 3 & 3 & 3 & 3 \\ \hline \end{array} + \begin{array}{|c|c|c|c|} \hline 1 & -1 & 1 & -1 \\ \hline 1 & -1 & 1 & -1 \\ \hline -1 & 1 & -1 & 1 \\ \hline -1 & 1 & -1 & 1 \\ \hline \end{array}$$

A
 $12 u_1 v_1\text{-transpose}$
 $4 u_2 v_2\text{-transpose}$

Rank-1 approximation: keep only the first piece, $A_1 = 3J$ (every entry 3, the average brightness). **Error:** $A - A_1$ is the second piece, whose entries are ± 1 , so its size is $\sqrt{16 \times 1} = 4 = \sigma_2$. **Energy kept:** $12^2 / (12^2 + 4^2) = 144 / 160 = 90\%$. **Storage:** A holds 16 numbers; A_1 needs u_1 , v_1 and $\sigma_1 = 4 + 4 + 1 = 9$. Savings grow fast for bigger matrices.

Example 3: Singular values reveal the rank

$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$; $A^T A = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix}$, trace 25, det $100 - 100 = 0$. $\lambda^2 - 25\lambda = 0$ gives $\lambda = 25$ and 0, so $\sigma = 5$ and 0. One nonzero singular value means rank 1: the plane is squashed onto a line. This agrees with $\det A = 1 \times 4 - 2 \times 2 = 0$ (Doc 6).

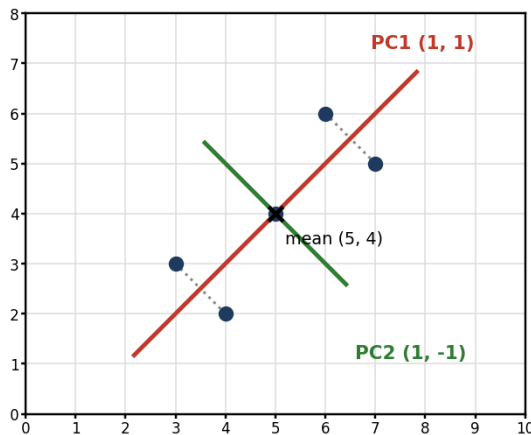
Quick reference card

SVD	$A = U \Sigma V^T$	U, V orthonormal; Σ = singular values, largest first
Recipe	$\sigma = \sqrt{\text{eigenvalues of } A^T A}$; v = its unit eigenvectors; $u = A v / \sigma$	check $A v = \sigma u$
Low rank	$A_k = \sigma_1 u_1 v_1^T + \dots + \sigma_k u_k v_k^T$	error = the discarded σ
PCA	center, $C = X_c^T X_c / (n - 1)$, eigenvectors	variance explained = $\lambda / \text{sum of } \lambda$
PCA and SVD	$\lambda_i = \sigma_i^2 / (n - 1)$	SVD of the centered data

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Part 2 of 3: Worked examples, continued

Example 4: PCA by hand on five data points



Data (x, y): (7, 5), (3, 3), (6, 6), (4, 2), (5, 4).

Step 1. Mean = $(25/5, 20/5) = (5, 4)$. Subtract it to get X_c : (2, 1), (-2, -1), (1, 2), (-1, -2), (0, 0).

Step 2. The grid gives $X_c^T X_c = \begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix}$. Divide by $n - 1 = 4$: $C = \begin{bmatrix} 2.5 & 2 \\ 2 & 2.5 \end{bmatrix}$.

Step 2: X_c -transpose times X_c

					2	1
					-2	-1
					1	2
					-1	-2
					0	0
2	-2	1	-1	0	10	8
1	-1	2	-2	0	8	10

Step 3. trace = 5, det = $6.25 - 4 = 2.25$. $\lambda^2 - 5\lambda + 2.25 = 0$ gives $\lambda = (5 \pm 4) / 2$: $\lambda = 4.5$ and 0.5 .

$\lambda = 4.5$: $\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$ gives direction **(1, 1)**. $\lambda = 0.5$: $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ gives **(1, -1)**. Unit vectors: divide by $\text{sqrt}(2) = 1.414$.

Step 5. Variance explained by PC1 = $4.5 / (4.5 + 0.5) = 90\%$. The dotted lines in the picture are the leftover distance, which is small.

SVD link: $X_c^T X_c$ has eigenvalues 18 and 2, so $\sigma = 4.243$ and 1.414 , and $\sigma^2 / 4 = 4.5$ and 0.5 .

Step 4. Scores (project onto each unit direction): PC1 score = $(x_c + y_c) / 1.414$, PC2 score = $(x_c - y_c) / 1.414$.

Point	Centered	PC1 score	PC2 score
(7, 5)	(2, 1)	2.121	0.707
(3, 3)	(-2, -1)	-2.121	-0.707
(6, 6)	(1, 2)	2.121	-0.707
(4, 2)	(-1, -2)	-2.121	0.707
(5, 4)	(0, 0)	0	0

Keeping only PC1 turns each 2D point into one number, and 90% of the variance survives. That is dimensionality reduction.

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Part 3 of 3: Practice. Use a pencil; yellow cells are yours. Name: _____ Date: _____

A. Orthonormal or not?

Pair of vectors	Dot product	Both length 1?	Orthonormal?
(0.6, 0.8) and (-0.8, 0.6)			
(1, 1) and (1, -1)			
(3, 4) and (4, -3)			
(1, 0) and (0, -1)			

For $U = \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix}$, compute $U^T U$. It should be the identity.

		0.6	-0.8
		0.8	0.6
0.6	0.8		
-0.8	0.6		

$U^T = \begin{bmatrix} 0.6 & 0.8 & -0.8 & 0.6 \end{bmatrix}$ is on the left. Each result cell is a dot product of two columns of U .

B. SVD from scratch: $A = \begin{bmatrix} 0 & 5 \\ -2 & 0 \end{bmatrix}$

Step 1 with the grid. Then fill the table. Remember to list σ from largest to smallest.

A-transpose times A

		0	5
		-2	0
0	-2		
5	0		

Step	Your answer
Eigenvalues of A-transpose A (big first)	
σ_1, σ_2	
v_1, v_2 (unit eigenvectors)	
$u_1 = A v_1 / \sigma_1, u_2 = A v_2 / \sigma_2$	

Check: write U, Σ, V^T in the grids and confirm the product gives A .

U times Σ

$(U \Sigma)$ times V-transpose

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Practice, continued

C. SVD of $A = \begin{bmatrix} 9 & 12 \\ -4 & 3 \end{bmatrix}$ (A -transpose A has eigenvalues 225 and 25)

Compute $A^T A$. Then $\sigma_1 = \sqrt{225}$ and $\sigma_2 = \sqrt{25}$. The unit eigenvectors are given as the columns of $V = \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix}$. The product $A V$ has columns $\sigma_i u_i$.

A -transpose times A

		9	12
		-4	3
9	-4		
12	3		

A times V

		0.6	-0.8
		0.8	0.6
9	12		
-4	3		

$\sigma_1 = \underline{\hspace{2cm}}$ $\sigma_2 = \underline{\hspace{2cm}}$

$u_1 = (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$ $u_2 = (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

(divide each column of $A V$ by its σ)

D. Low-rank approximation of a 4×4 matrix

A has $\sigma_1 = 8$, $\sigma_2 = 4$, $u_1 = v_1 = \frac{1}{2}(1, 1, 1, 1)$, $u_2 = \frac{1}{2}(1, 1, -1, -1)$, $v_2 = \frac{1}{2}(1, -1, -1, 1)$. Compute each piece $\sigma u v^T$: every entry is $\sigma \times (\text{entry of } u) \times (\text{entry of } v)$.

8 $u_1 v_1$ -transpose

+

4 $u_2 v_2$ -transpose

=

3	1	1	3
3	1	1	3
1	3	3	1
1	3	3	1

A (given)

Do the two pieces add up to A ? $\underline{\hspace{2cm}}$. The best rank-1 approximation is piece number $\underline{\hspace{2cm}}$.

Size of the error $A - A_1 = \sigma_2 = \underline{\hspace{2cm}}$. Energy kept = $8^2 / (8^2 + 4^2) = \underline{\hspace{2cm}}\%$. Numbers to store A_1 : $4 + 4 + 1 = \underline{\hspace{2cm}}$.

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Practice, continued

E. PCA by hand

Data (x, y): (7, 6), (1, 4), (5, 8), (3, 2), (4, 5). Follow the five steps of PCA.

Steps 1-2. The mean is (__ , __). Write the centered data into the grid twice: as X_c^T (2 rows, 5 columns) on the left, and as X_c (5 rows, 2 columns) on top. Then multiply.

X_c -transpose times X_c

Step 3. $C = (\text{grid}) / 4 =$

trace = ____, det = ____, so $\lambda_1 =$ ____, $\lambda_2 =$ ____.

Eigenvector for λ_1 : (__ , __). For λ_2 : (__ , __).

Step 5. Variance explained by PC1 = ____ / ____ = ____ %.

Step 4. Scores: $PC1 = (x_c + y_c) / 1.414$ and $PC2 = (x_c - y_c) / 1.414$, using your directions (1, 1) and (1, -1).

Point	Centered	PC1 score	PC2 score
(7, 6)			
(1, 4)			
(5, 8)			
(3, 2)			
(4, 5)			

F. PCA and SVD are the same thing

Quantity	First direction	Second direction
Eigenvalues of X_c -transpose X_c (from the grid)		
Singular values $\sigma = \text{sqrt}(\text{eigenvalue})$, 3 decimals		
$\sigma^2 / (n - 1)$ (should equal the eigenvalues of C)		

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Practice, continued

G. True or False?

Statement	True / False
1. Singular values are never negative.	
2. The rank of a matrix equals its number of nonzero singular values.	
3. The best rank-k approximation keeps the k largest singular values.	
4. The principal directions are eigenvectors of the covariance matrix.	
5. PCA works best if you skip centering the data.	
6. A 2×2 matrix with second singular value 0 squashes the plane onto a line.	

H. Image compression: how many numbers?

A 100×100 grayscale image has 10,000 numbers. A rank-k approximation stores $k(100 + 100 + 1) = 201k$ numbers.

Rank k	Numbers stored = $201k$	% of the original 10,000
1		
5		
10		
20		
50		

Above what rank k does the low-rank form stop saving space? $k = \underline{\hspace{2cm}}$ (hint: solve $201k < 10,000$).

I. Reflection

In two sentences, explain why keeping only the largest singular values gives a good compressed image.

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Answer key (page 1 of 2): check your work only after finishing

A. Orthonormal or not?

Pair of vectors	Dot product	Both length 1?	Orthonormal?
(0.6, 0.8) and (-0.8, 0.6)	0	Yes	Yes
(1, 1) and (1, -1)	0	No	No
(3, 4) and (4, -3)	0	No	No
(1, 0) and (0, -1)	0	Yes	Yes

		0.6	-0.8
		0.8	0.6
0.6	0.8	1	0
-0.8	0.6	0	1

$U^T U = \text{identity}$. (1, 1) and (1, -1) are perpendicular but not unit length; (3, 4) has length 5.

B. SVD of $A = \begin{bmatrix} 0 & 5 \\ -2 & 0 \end{bmatrix}$

A-transpose A

		0	5
		-2	0
0	-2	4	0
5	0	0	25

$U \Sigma$

		5	0
		0	2
1	0	5	0
0	-1	0	-2

$(U \Sigma) V\text{-transpose}$

		0	1
		1	0
5	0	0	5
0	-2	-2	0

$A^T A = \begin{bmatrix} 4 & 0 \\ 0 & 25 \end{bmatrix}$: eigenvalues 25, 4, so $\sigma = 5, 2$. Eigenvectors (0, 1) and (1, 0), so $V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. $u_1 = A(0, 1) / 5 = (1, 0)$ and $u_2 = A(1, 0) / 2 = (0, -1)$. The product returns A.

C. SVD of $A = \begin{bmatrix} 9 & 12 \\ -4 & 3 \end{bmatrix}$

A-transpose A

		9	12
		-4	3
9	-4	97	96
12	3	96	153

$A V$

		0.6	-0.8
		0.8	0.6
9	12	15	0
-4	3	0	5

$\sigma = 15$ and 5. $A V = \begin{bmatrix} 15 & 0 \\ 0 & 5 \end{bmatrix}$, so $u_1 = (15, 0) / 15 = (1, 0)$ and $u_2 = (0, 5) / 5 = (0, 1)$: U is the identity and $A = \Sigma V^T$.

D. Low-rank approximation

$$\begin{bmatrix} 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 1 & 3 \\ 3 & 1 & 1 & 3 \\ 1 & 3 & 3 & 1 \\ 1 & 3 & 3 & 1 \end{bmatrix}$$

$8 u_1 v_1^T = 2J$ $4 u_2 v_2^T$ A

Yes, they add up to A. Best rank-1 approximation: piece 1 (every entry 2). Error size = 4. Energy = $64 / 80 = 80\%$. Numbers to store = 9.

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Answer key (page 2 of 2)

E. PCA by hand

$X_c^T X_c$

					3	1
					-3	-1
					1	3
					-1	-3
					0	0
3	-3	1	-1	0	20	12
1	-1	3	-3	0	12	20

Mean = (4, 5). Grid = [20 12; 12 20], so $C = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$.

trace = 10, det = 25 - 9 = 16. $\lambda^2 - 10\lambda + 16 = 0$ gives $\lambda = 8, 2$.

$\lambda = 8$: $\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix}$ gives **(1, 1)**. $\lambda = 2$: $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$ gives **(1, -1)**.

Variance explained by PC1 = 8 / 10 = **80%**.

Point	Centered	PC1 score	PC2 score
(7, 6)	(3, 1)	2.828	1.414
(1, 4)	(-3, -1)	-2.828	-1.414
(5, 8)	(1, 3)	2.828	-1.414
(3, 2)	(-1, -3)	-2.828	1.414
(4, 5)	(0, 0)	0	0

PC1 = $(x_c + y_c) / 1.414$: e.g. $(3 + 1) / 1.414 = 2.828$. PC2 = $(x_c - y_c) / 1.414$: e.g. $(3 - 1) / 1.414 = 1.414$.

F. PCA and SVD

$X_c^T X_c = \begin{bmatrix} 20 & 12 \\ 12 & 20 \end{bmatrix}$ has eigenvalues 32 and 8 (they are 4 × the eigenvalues of C). $\sigma = \sqrt{32} = 5.657$ and $\sqrt{8} = 2.828$. $\sigma^2 / 4 = 8$ and 2: the eigenvalues of C.

G. True or False?

1 True. 2 True. 3 True. 4 True. 5 **False** (always center first). 6 True (rank 1: the plane is squashed onto a line).

H. Image compression

Rank k	Numbers stored	% of 10,000
1	201	2.01%
5	1005	10.05%
10	2010	20.1%
20	4020	40.2%
50	10050	100.5% (no saving)

201 $k < 10,000$ gives $k < 49.75$, so the saving stops above $k = 49$. In real images a few large singular values usually carry most of the picture; the many tiny ones are mostly fine detail or noise.

I. Reflection

A model answer: the singular values are sorted by importance, and the biggest ones capture most of the image's energy (its broad shapes). Dropping the tiny ones loses only fine detail or noise while storing far fewer numbers.

Key takeaways: $A = U \Sigma V^T$; $\sigma = \sqrt{\text{eigenvalues of } A^T A}$; rank = number of nonzero σ ; the best rank-k approximation keeps the top k pieces; PCA = center, covariance, eigenvectors, project; $\lambda = \sigma^2 / (n - 1)$.

Next: Doc 10 - Derivatives and partial derivatives (Module 2: Calculus).